Veille informatique quantique - 2026-10-23

Période analysée : les 7 derniers jours. Articles retenus : 10.

Synthèse

1. Multi-FPGA Synchronization and Data Communication for Quantum Control and Measurement

Source : arXiv Quantum Physics - publié le 2026-10-23

Scores : pertinence 87/100 | nouveauté 50/100 | importance 50/100 | score 67/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2506.09856v2 Announce Type: replace Abstract: Quantum computing has grown from novel physics experiments with a few qubits controlled by a single controller to systems with qubit counts that require multiple controllers working in concert. While a few expensive multi-board commercial solutions exist, most open-source solutions are limited to single-board radio frequency system-on-chip (RFSoC). The essential requirements for a multi-board solution are clock synchronization among multiple b

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computing, qubit, quantum processor, quantum algorithm.

2. Reducing QPU Calls in Variational Quantum Algorithms via Classical Offloading of Gradient Computation

Source : arXiv Quantum Physics - publié le 2026-10-23

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2503.16361v3 Announce Type: replace Abstract: Variational quantum algorithms (VQAs) are a promising approach for leveraging near-term quantum computers, but trainability and quantum-resource costs remain major challenges. Here we address these challenges with the \textbf{H}ardware \textbf{E}fficient and dynamical \textbf{LI}e algebra supported \textbf{A}nsatz (HELIA) and two training methods combining the classical-enhanced \g method with the quantum-based Parameter-Shift Rule (PSR). This

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Mots-clés détectés : quantum computer, qubit, quantum algorithm.

3. Fault-Tolerant Logical Operations and Efficient State Preparation in Modular Quantum Architectures with Noisy Interfaces

Source : arXiv Quantum Physics - publié le 2026-10-23

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2607.27204v2 Announce Type: replace Abstract: Modular quantum computing is a leading paradigm for scaling quantum computation beyond the resource limitations of monolithic devices. In this architecture, multiple quantum processing units (QPUs), employing identical or distinct qubit modalities, are interconnected via shared entanglement. Here, we investigate how errors at module interfaces and within individual QPUs affect fault-tolerant computation when qubits are encoded using the rotate

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Mots-clés détectés : quantum computing, qubit, quantum error correction.

4. Normal-Form Characterization of Adaptivity and Entanglement in Pauli-Based Computation

Source : arXiv Quantum Physics - publié le 2026-10-23

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.30315v1 Announce Type: new Abstract: Pauli-based computation (PBC) is a measurement-based model of quantum computation that samples from output distributions of universal Clifford+T circuits. Here, we provide a normal form (NF) that completely characterizes its adaptive and entanglement structures. These insights unify PBC with one-way quantum computing as dual or `time-reversed' counterparts. We use the NF to advance strategies for distributed computation and investigate magic-sta

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computing, qubit.

5. High-fidelity and robust controlled-Z gates for superconducting transmon qubits. II. System-parameter uncertainties and overall errors

Source : arXiv Quantum Physics - publié le 2026-10-23

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.30681v1 Announce Type: new Abstract: In the companion paper (Part I), we developed an optimal-control framework to suppress intrinsic and decoherence-induced errors in controlled-Z (CZ) gates between directly coupled, frequency-tunable transmons. Here, we extend this framework to incorporate uncertainties in coupling strength, anharmonicities, and zero-flux qubit frequencies, enabling the design of high-fidelity gates robust against noise and parameter uncertainties. We classify thes

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : qubit, quantum processor.

6. Noise redistribution by stabilizer codes: an organizing view of concatenated three-qubit codes

Source : arXiv Quantum Physics - publié le 2026-10-23

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.31007v1 Announce Type: new Abstract: We present an organizing framework for concatenated quantum error correction based on the observation that the syndrome projection of a stabilizer code reshapes the noise channel in a code-determined manner. For any stabilizer code, one level of QEC maps the physical noise to a block-diagonal logical channel whose diagonal Pauli weights are fixed by the code geometry. Coherences between different syndrome classes vanish, and no physical Pauli twir

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : qubit, quantum error correction.

7. Norm-Aware Product Formulas for Hamiltonian Simulations

Source : arXiv Quantum Physics - publié le 2026-10-23

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.31070v1 Announce Type: new Abstract: With the target of efficiently solving the quantum-mechanical Schr\"odinger equation on future quantum computers, first- and second-order product formulas for the time-evolution operator are proposed, designed to exploit hierarchies in the operator norms of the individual Hamiltonian components. Inspired by the THRIFT product formula, these decomposition schemes achieve a suppression of the leading error term by a factor of alpha<1 with respect to

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computer, qubit.

8. Automating quantum circuit optimization with AI agents

Source : arXiv Quantum Physics - publié le 2026-10-23

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2610.31352v1 Announce Type: new Abstract: Compiling quantum algorithms for real devices involves hard optimization problems that can decide whether an algorithm runs at scale. We present a general method for solving such problems with AI agents, with every result certified. It builds on autoresearch loops, in which agents iteratively improve programs against a fixed evaluator and which have recently improved quantum cryptanalysis circuits, and adds two elements: a frozen, exact verifier c

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Mots-clés détectés : qubit, quantum algorithm.

9. Meta-learning of Gibbs states for many-body Hamiltonians with applications to Quantum Boltzmann Machines

Source : arXiv Quantum Physics - publié le 2026-10-23

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2507.16373v3 Announce Type: replace Abstract: The preparation of quantum Gibbs states is a fundamental challenge in quantum computing, essential for applications ranging from modeling open quantum systems to quantum machine learning. Building on the Meta-Variational Quantum Eigensolver framework proposed by Cervera-Lierta et al. and a problem-driven ansatz design, we introduce two meta-learning algorithms: Meta-Variational Quantum Thermalizer (Meta-VQT) and Neural Network Meta-VQT (NN-Met

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Mots-clés détectés : quantum computing, qubit.

10. Space-Time Optimisations for Early Fault-Tolerant Quantum Computation

Source : arXiv Quantum Physics - publié le 2026-10-23

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2511.08848v2 Announce Type: replace Abstract: Fault-tolerance is the future of quantum computing, ensuring error-corrected quantum computation that can be used for practical applications. Resource requirements for fault-tolerant quantum computing (FTQC) are daunting, and hence, compilation techniques must be designed to ensure resource efficiency. There is a growing need for compilation strategies tailored to the early FTQC regime, which refers to the first generation of fault-tolerant ma

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Méthode

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